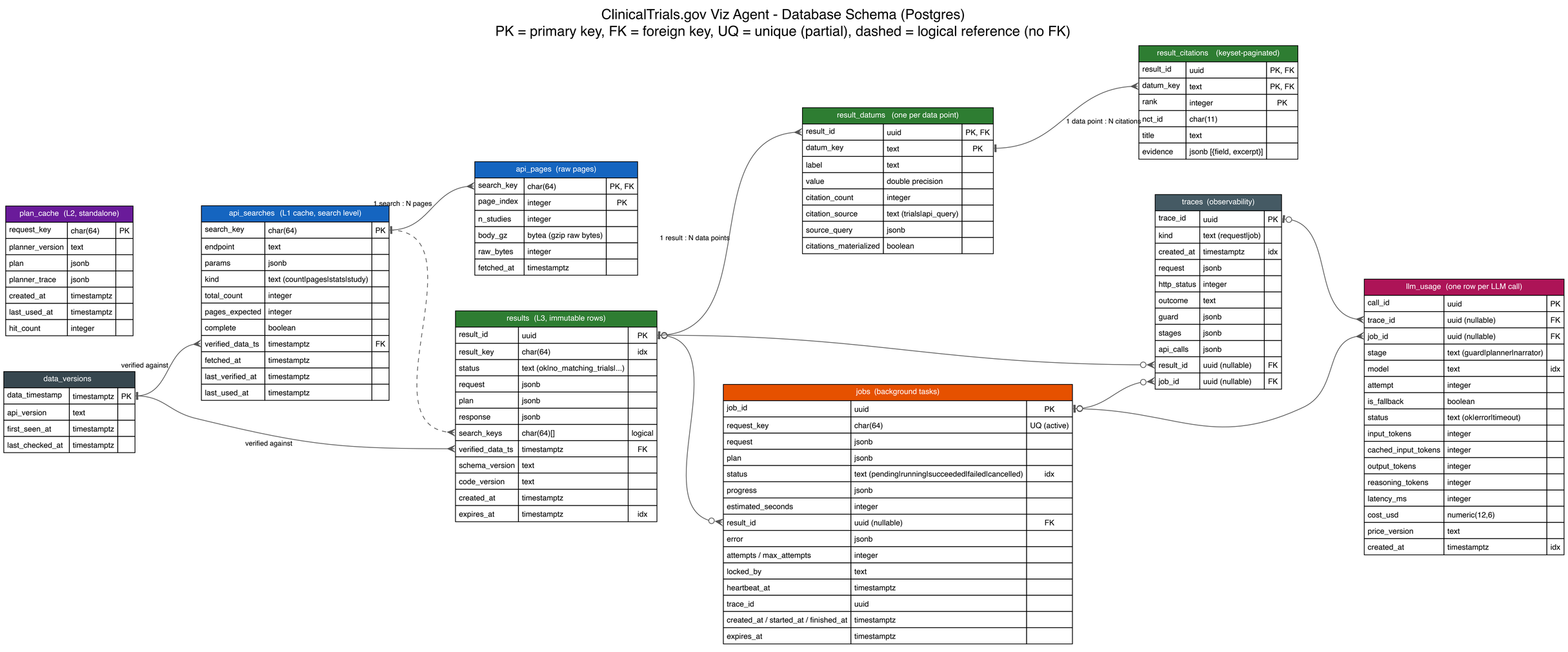
ClinicalTrials.gov Viz Agent - Database Schema (Postgres)
PK = primary key, FK = foreign key, UQ = unique (partial), dashed = logical reference (no FK)
plan_cache (L2, standalone)
request_key char(64) PK
planner_version text
plan jsonb
planner_trace jsonb
created_at timestamptz
last_used_at timestamptz
hit_count integer
data_versions
data_timestamp timestamptz PK
api_version text
first_seen_at timestamptz
last_checked_at timestamptz
verified against
verified against
api_searches (L1 cache, search level)
search_key char(64) PK
endpoint text
params jsonb
kind text (countlpageslstatslstudy)
total_count integer
pages_expected integer
complete boolean
verified_data_ts timestamptz FK
fetched_at timestamptz
last_verified_at timestamptz
last_used_at timestamptz
1 search : N pages
api_pages (raw pages)
search_key char(64) PK, FK
page_index integer PK
n_studies integer
body_gz bytea (gzip raw bytes)
raw_bytes integer
fetched_at timestamptz
results (L3, immutable rows)
result_id uuid PK
result_key char(64) idx
status text (oklno_matching_trialsl...)
request jsonb
plan jsonb
response jsonb
search_keys char(64)[] logical
verified_data_ts timestamptz FK
schema_version text
code_version text
created_at timestamptz
expires_at timestamptz idx
1 result : N data points
result_datums (one per data point)
result_id uuid PK, FK
datum_key text PK
label text
value double precision
citation_count integer
citation_source text (trialslapi_query)
source_query jsonb
citations_materialized boolean
1 data point : N citations
result_citations (keyset-paginated)
result_id uuid PK, FK
datum_key text PK, FK
rank integer PK
nct_id char(11)
title text
evidence jsonb [{field, excerpt}]
jobs (background tasks)
job_id uuid PK
request_key char(64) UQ (active)
request jsonb
plan jsonb
status text (pendinglrunninglsucceededlfailedlcancelled) idx
progress jsonb
estimated_seconds integer
result_id uuid (nullable) FK
error jsonb
attempts / max_attempts integer
locked_by text
heartbeat_at timestamptz
trace_id uuid
created_at / started_at / finished_at timestamptz
expires_at timestamptz
traces (observability)
trace_id uuid PK
kind text (requestljob)
created_at timestamptz idx
request jsonb
http_status integer
outcome text
guard jsonb
stages jsonb
api_calls jsonb
result_id uuid (nullable) FK
job_id uuid (nullable) FK
llm_usage (one row per LLM call)
call_id uuid PK
trace_id uuid (nullable) FK
job_id uuid (nullable) FK
stage text (guardlplannerlnarrator)
model text idx
attempt integer
is_fallback boolean
status text (oklerrorltimeout)
input_tokens integer
cached_input_tokens integer
output_tokens integer
reasoning_tokens integer
latency_ms integer
cost_usd numeric(12,6)
price_version text
created_at timestamptz idx